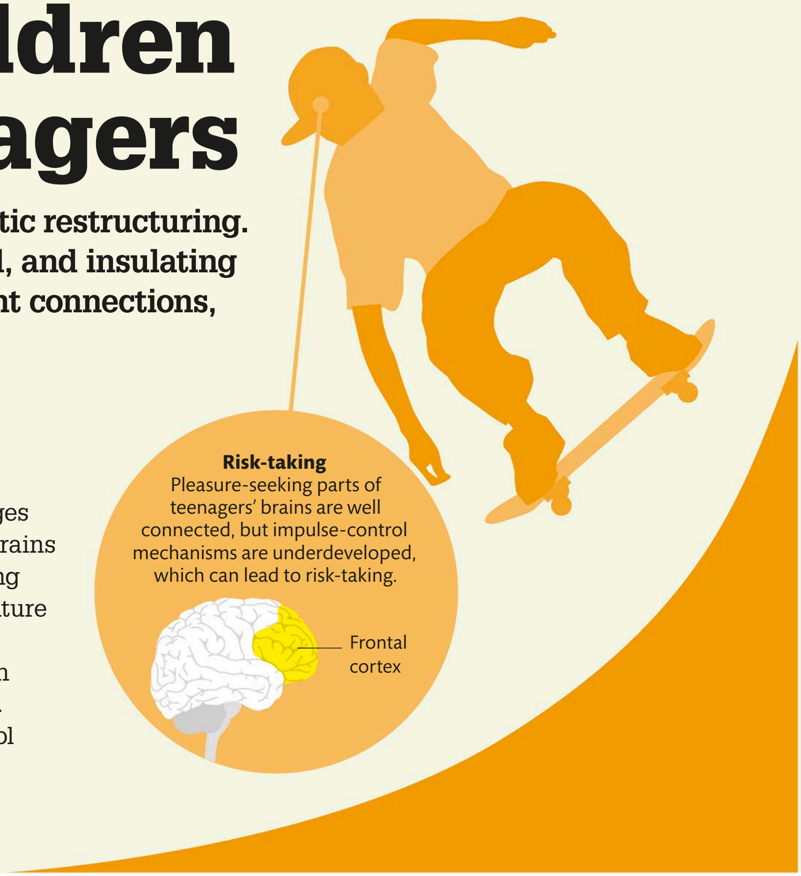
Older Children and Teenagers

Teenage brains undergo dramatic restructuring. Unused connections are pruned, and insulating myelin coats the most important connections, making them more efficient.

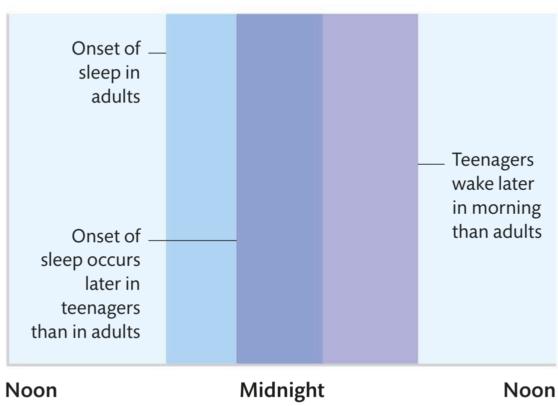
Teenage behavior

Teenagers have a reputation for being impulsive, rebellious, self-centered, and emotional. A lot of this is due to the changes happening in adolescent brains. Human brains change and develop in set patterns, leaving teenagers with a mix of mature and immature brain regions as they grow. The last area to fully develop is the frontal cortex, which regulates the brain and controls impulses. This area allows adults to exert self-control over their emotions and desires, which is something adolescents can struggle with.



Sleep cycles

During our teenage years, we need plenty of sleep as our brain continues to develop. But at this time, our circadian rhythms shift as melatonin, the hormone that is released in the evening and makes us feel sleepy, begins to be released later than usual. This is why teenagers often want to go to bed later than children and adults and may struggle to get up for school in the morning.



KEY
- Adult sleep time
- Adolescent sleep time

Out of sync
Waking teenagers early for school is like giving them constant jet lag. Studies have shown that starting school an hour later improved attendance and grades. Fights and even car accidents also decreased.

SYNAPTIC PRUNING

Synaptic pruning, which is when unused neural connections die off, starts during childhood and continues through our teen years. Cortical areas are pruned from the back to the front. Pruning makes each area more efficient, so until it is finished, that region cannot be considered fully mature.

IMMATURE MATURE